RAMDEOBABA UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



B. TECH PROJECT REPORT

B.Tech. in Computer Science and Engineering (Artificial Intelligence and Cyber Security)

Idea Lab

ON

Multimodal AI-Based Machine Health Monitoring & Predictive Maintenance

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Under the Guidance of
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Academic Year: 2026 – 27

RAMDEOBABA UNIVERSITY, NAGPUR

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CERTIFICATE

This is to certify that the Project Report on "**Multimodal AI-Based Machine Health Monitoring & Predictive Maintenance**" is a bonafide work of Vaishnav Balpande, Shreyank Nipane, Tanushri Sapate, Aastha Pusdekar and Kartik Madankar, submitted to Ramdeobaba University, Nagpur in partial fulfilment of the award of a Degree of Bachelor of Technology (B.Tech), in Computer Science and Engineering (Artificial Intelligence and Cyber Security). It has been carried out at the Department of Computer Science and Engineering, Ramdeobaba University, Nagpur during the academic year 2026-27.

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STUDENT DECLARATION

We hereby declare that the project report entitled "**Multimodal AI-Based Machine Health Monitoring & Predictive Maintenance**" is submitted as part of the requirements for the B.Tech programme at Ramdeobaba University.

The work presented in this report is prepared by the undersigned students under the guidance of the project supervisor. All external sources, datasets, software tools, figures and technical material used in the project have been appropriately acknowledged. The prototype currently operates on prerecorded public datasets that simulate real sensor inputs, and the results reported here should be read in that context.

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This report entitled "**MULTIMODAL AI-BASED MACHINE HEALTH MONITORING & PREDICTIVE MAINTENANCE**" by **VAISHNAV BALPANDE, SHREYANK NIPANE, TANUSHRI SAPATE, AASTHA PUSDEKAR AND KARTIK MADANKAR** is approved for the degree of Bachelor of Technology (B.Tech).

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We acknowledge the support of our classmates, laboratory staff, friends and family members who contributed directly or indirectly to the completion of this work.

ABSTRACT

Industrial machines develop faults through gradual changes in temperature, vibration and acoustic behaviour long before a visible or catastrophic failure occurs. Most existing condition-monitoring systems analyse these signals independently, leaving operators and maintenance teams with fragmented information rather than a single machine-health picture.

This project presents a multimodal, edge-oriented machine health monitoring and predictive maintenance platform that combines thermal, vibration and acoustic evidence. Each modality is processed using domain-specific signal analysis -- thermal frame analysis, windowed vibration features such as RMS, kurtosis, crest factor and FFT-based spectral features, and acoustic features such as STFT, octave bands and MFCC. Modality-specific models (MobileNetV3-Small for thermal imagery, Random Forest for vibration and Logistic Regression for acoustic signals) analyse their inputs and output predictions with confidence scores. Their outputs are displayed as separate evidence in the prototype. Although the software includes health-engine and evidence-fusion components, the available demonstration recordings are not verified as synchronized measurements of the same machine; therefore, the project does not claim a validated fused diagnosis across all three modalities. A deterministic health engine with baseline calibration, z-score detection, Isolation Forest and temporal smoothing remains the authoritative source of machine health.

The system is implemented with a Python and FastAPI backend, WebSocket-based live telemetry, SQLite persistence and a React and TypeScript operator dashboard that presents a machine-health score on a 0-100 scale, where available; this score is not a percentage probability. The dashboard also shows current state, multimodal evidence, events and history. The prototype supports three sensing modalities, three machine-learning models and a demonstration set of seven usable videos covering six machine types, and has passed 194 backend unit, integration and API tests in the reported test run.

Keywords: Predictive Maintenance, Multimodal Sensing, Machine Health Monitoring, Anomaly Detection, Evidence Fusion, Thermal Imaging, Vibration Analysis, Acoustic Analysis, Edge Computing

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LIST OF ABBREVIATIONS

Table: List of Abbreviations

Abbreviation	**Meaning**
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DAQ	Data Acquisition
FFT	Fast Fourier Transform
IoT	Internet of Things
MFCC	Mel-Frequency Cepstral Coefficients
ML	Machine Learning
RMS	Root Mean Square
RUL	Remaining Useful Life
STFT	Short-Time Fourier Transform
UI	User Interface

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

Industrial machines such as motors, engines, hydraulic systems, transformers and battery packs are central to manufacturing and infrastructure. Their failure can lead to unplanned downtime, production loss, maintenance costs and equipment damage. Condition monitoring aims to detect developing faults early so that maintenance can be planned rather than reactive.

1.1 Background

Machine faults commonly manifest through changes in temperature, vibration and acoustic behaviour before any visible or catastrophic failure. Vibration signals are effective for detecting mechanical and bearing-related abnormalities, acoustic signatures capture sound patterns associated with machine operation, and thermal imaging reveals hotspots, thermal gradients and abnormal heating caused by friction or overheating.

Advances in signal processing, machine learning and low-cost edge computing now make it feasible to analyse these signals locally and in real time. A system that brings multiple physical indicators together can provide a more comprehensive basis for early anomaly detection and predictive maintenance than any single sensor.

1.2 Problem Statement

Industrial machines can develop faults through changes in temperature, vibration and acoustic behaviour before visible or catastrophic failure. Existing monitoring systems often analyse these signals independently. This project demonstrates separate thermal, vibration and acoustic analysis paths in one interface. Because the available recordings are not verified as synchronized measurements of the same machine, the prototype does not claim a validated unified diagnosis.

Who is affected

- Industrial operators -- need continuous visibility of machine condition.
- Maintenance teams -- need earlier indication of developing abnormalities.
- Plant managers -- face unplanned downtime and maintenance costs.
- Equipment owners -- need scalable monitoring across multiple machines.

Current challenges

- Single-sensor monitoring provides only partial information about machine condition.
- Raw vibration, acoustic and thermal data require specialised signal-processing expertise.
- Different fault indicators may appear across different sensing modalities.
- Conventional systems often provide separate measurements and alarms rather than unified evidence.

Why the problem is important

Machine failures can lead to unplanned downtime, production loss, maintenance costs and equipment damage. A system capable of continuously combining multiple physical indicators can provide a more comprehensive basis for early anomaly detection and predictive maintenance.

1.3 Objectives

- Acquire and process data: collect and process thermal, vibration and acoustic machine data while extracting relevant features.
- Detect anomalies: analyse vibration, acoustic and thermal patterns to identify abnormal machine behaviour.
- Fuse evidence: implement a weighted evidence-fusion component and assess its output once synchronized multimodal data is available.
- Build a monitoring dashboard: display telemetry, machine health, events, alerts and system status from recorded inputs in a unified interface.

Measurable system goals

- Real-time telemetry and health updates.
- Sensor analysis with missing-data handling.
- Event and history visualisation.
- Support for demonstration (dataset-driven) and live data.

1.4 Scope

The project covers multimodal data acquisition, modality-specific signal processing and feature extraction, machine-learning inference, confidence-aware evidence fusion, a deterministic health engine, telemetry persistence and a real-time operator dashboard. The current prototype is dataset-driven: prerecorded thermal video, vibration and

acoustic datasets simulate real sensor inputs. It is intended for academic and engineering demonstration and should not be treated as a certified industrial safety system until validated on physical hardware and plant conditions.

1.5 Organization of the Report

Chapter 1 introduces the problem and objectives. Chapter 2 presents the literature review and research gap. Chapter 3 describes the proposed system and methodology. Chapter 4 discusses system design and implementation. Chapter 5 presents results and discussion. Chapter 6 concludes the work and presents the innovation, impact, deliverables and future scope.

CHAPTER 2

LITERATURE REVIEW AND RESEARCH GAP

2. LITERATURE REVIEW AND RESEARCH GAP

2.1 Introduction

Machine condition monitoring has traditionally been approached through individual sensing modalities. This chapter reviews vibration-based, acoustic, thermal and conventional single-modality approaches, summarises their strengths and limitations, and identifies the research gap addressed by this project.

2.2 Existing Approaches

2.2.1 Vibration-based monitoring

Vibration-based monitoring uses accelerometers to monitor rotating-machine vibration. Common analysis includes RMS, kurtosis, crest factor, FFT and spectral analysis. It is effective for detecting mechanical and bearing-related abnormalities; however, it primarily represents mechanical behaviour and may miss thermal or acoustic indicators.

2.2.2 Acoustic / sound monitoring

Acoustic monitoring uses microphones to capture machine acoustic signatures. Analysis includes FFT, STFT, octave / 1/3-octave bands and MFCC, and can identify abnormal sound patterns associated with machine operation. Its limitation is sensitivity to environmental noise, operating conditions and recording location.

2.2.3 Thermal / infrared monitoring

Thermal monitoring uses thermal imaging to identify hotspots, thermal gradients and abnormal heating. It is useful for observing friction, overheating and temperature-pattern changes, and provides spatial information unavailable from point sensors. However, thermal inspection alone cannot directly characterise vibration or acoustic faults.

2.2.4 Conventional single-modality systems

Conventional systems typically monitor one sensor type independently and generate separate alarms, measurements or inspection reports. They require operators to interpret multiple sources manually and offer no unified interpretation of thermal, vibration and acoustic evidence.

2.3 Comparison of Existing Approaches

Table 2.1: Comparison of Existing Monitoring Approaches

Approach	Sensing / Analysis	Strength	Limitation
Vibration-based	Accelerometers; RMS, kurtosis, crest factor, FFT	Detects mechanical and bearing-related abnormalities	Primarily mechanical; may miss thermal or acoustic indicators
Acoustic / sound	Microphones; FFT, STFT, octave / 1/3-octave, MFCC	Identifies abnormal sound patterns of machine operation	Sensitive to environmental noise, operating conditions and recording location
Thermal / infrared	Thermal imaging; hotspots and gradients	Observes friction, overheating; gives spatial information	Cannot directly characterise vibration or acoustic faults
Conventional single-modality	One sensor type, analysed independently	Simple, familiar to operators	Separate alarms; no unified thermal + vibration + acoustic interpretation
Proposed system	Thermal + vibration + acoustic with AI inference	Confidence-aware evidence fusion (prototype)	Prototype evaluated on recorded datasets; synchronized validation and physical plant pilot pending

2.4 Limitations of Existing Work

Existing approaches analyse machine condition mainly through individual sensor modalities. Each modality captures only part of the machine state, specialised signal-processing expertise is needed to interpret raw data, and alarms from different sources are not reconciled. Environmental noise further reduces the reliability of single-sensor interpretations.

2.5 Research Gap

There is a need for a lightweight, edge-oriented system that combines thermal, vibration and acoustic evidence with signal processing, AI-based inference and confidence-aware fusion to provide a unified machine-health view. The proposed system addresses this gap by preserving domain-specific signal features before machine learning, fusing normalised evidence rather than forcing alignment of raw feeds, and gating uncertain predictions to avoid alert fatigue.

CHAPTER 3

PROPOSED SYSTEM AND METHODOLOGY

3. PROPOSED SYSTEM AND METHODOLOGY

3.1 Proposed System

The proposed system processes vibration, acoustic and thermal inputs through modality-specific analysis paths and presents the available results in a common monitoring interface. The prototype demonstrates these paths using recorded datasets. It has not established that combining these datasets improves diagnostic accuracy, because the recordings are not verified as synchronized measurements of the same machine. The system operates in four stages: machine sensors, process and analyse, combine the evidence, and display to the user.

3.2 System Architecture

The architecture is organised into four layers: (1) input / data acquisition through multimodal sensing, (2) signal processing and feature extraction, (3) AI / anomaly intelligence combining deterministic and machine-learning methods, and (4) output / monitoring that provides an actionable machine-health view.

3.3 Data Acquisition

Three modalities are acquired. Thermal data is frame-by-frame thermal imagery from a thermal camera (dataset-driven in the demonstration). Vibration data comes from a vibration sensor or the IMS bearing dataset as windowed signals sampled at 20 kHz with 20,480 samples per window. Acoustic data comes from a microphone or an acoustic dataset as audio recordings of 10 s at 16 kHz. In the prototype, machine, sensors and datasets feed an edge-processing stage.

3.4 Signal Processing and Feature Extraction

Each modality is analysed with domain-specific methods so that the physical characteristics of the signal are preserved before machine learning. The output of this stage is cleaned signals and modality-specific health features.

Table 3.1: Modality-Specific Analysis and Features

Modality	Processing steps	Features extracted
Thermal	Frame preprocessing; hotspot detection	Relative intensity, hotspot area and centroid, thermal gradients, hotspot evolution
Vibration	Sliding-window analysis	RMS, kurtosis, crest factor, FFT and spectral features, spectral centroid / bandwidth, octave / 1/3-octave energy
Acoustic	Audio preprocessing; FFT and STFT	Octave / 1/3-octave bands, MFCC, spectral features

For a window of n samples x_1 ... x_n, two of the time-domain features used for vibration are:

$$\text{RMS} = \text{sqrt}((1/n) * \text{sum } x_i^2), \quad \text{Crest Factor} = \max|x| / \text{RMS}$$

3.5 AI and Anomaly Intelligence

3.5.1 Deterministic health engine

The deterministic health engine performs baseline calibration, statistical z-score detection, Isolation Forest anomaly detection, severity estimation and temporal smoothing with hysteresis. It remains the authoritative source of machine health; machine-learning outputs act as supplementary evidence. For a feature value x with baseline mean mu and standard deviation sigma, the z-score is:

$$z = (x - \mu) / \sigma$$

3.5.2 Modality-specific machine learning models

Table 3.2: Machine Learning Models Used

Modality	Model	Remarks
Thermal	MobileNetV3-Small (CNN)	Transfer learning; edge-optimised
Vibration	Random Forest	Trained on vibration window features
Acoustic	Logistic Regression (scikit-learn)	Trained on 61 acoustic features

3.5.3 Confidence handling and evidence combination

The system can suppress or qualify low-confidence model outputs according to its configured thresholds. The dashboard displays modality-level predictions and confidence where available. A weighted-fusion formula may be used by the software components, but the current demonstration does not establish a validated combined

diagnosis because its modality recordings are not verified as synchronized measurements of the same machine. The deterministic health assessment and ML evidence should be reported separately wherever the demo does not expose a verified combined result.

3.6 Output and Monitoring

The output stage presents a machine-health score on a 0-100 scale, where available (not a percentage probability), the current machine state, multimodal evidence (prediction and confidence per modality), events and history, and an operator interface in the form of a monitoring dashboard.

Table 3.3: Machine Health States

State	Meaning
Normal	Machine health is within the calibrated baseline
Warning	Deviation from baseline detected; early indication of a developing abnormality
Critical	Significant anomaly detected; maintenance attention required
Unknown	Insufficient or low-confidence evidence to assess machine health

3.7 Evaluation Approach

The prototype is evaluated through functional and software-quality criteria: correct operation of each modality pipeline, handling of missing modalities, real-time telemetry and health updates, event and history persistence, and automated testing of the backend through unit, integration and API tests.

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

4. SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Architecture

The implementation is divided into sensing, edge processing, storage and visualisation modules. Sensors (thermal camera, vibration sensor and microphone) feed an edge computer that performs local processing, inference and fusion. During the prototype stage, prerecorded datasets simulate the real sensor inputs. Technologies are grouped into four areas: backend and real-time, monitoring dashboard, machine learning, and signal processing. The system itself comprises a multimodal pipeline, a health engine, and telemetry and persistence.

4.2 Hardware and Data Sources

Table 4.1: Hardware and Data Sources

Component	Purpose	Prototype source
Thermal camera	Thermal imagery and hotspot monitoring	Thermal video / image datasets
Vibration sensor	Rotating-machine vibration acquisition	IMS bearing dataset
Microphone	Machine acoustic signatures	Bearing acoustic dataset (DCASE-style)
Edge computer	Local processing, inference and fusion	Development computer

The prototype environment uses prerecorded datasets that simulate real sensor inputs; direct hardware data acquisition is planned as future work.

4.3 Software Components

Table 4.2: Software Technologies Used

Area	Technologies
Backend and real-time	Python, FastAPI, SQLite, WebSocket
Monitoring dashboard	React, TypeScript, Vite
Machine learning	PyTorch (thermal); scikit-learn (vibration and acoustic)
Signal processing	NumPy, SciPy, librosa, OpenCV

4.4 Implementation Modules

4.4.1 Multimodal pipeline

- Real-time processing of incoming data.
- Thermal frame analysis.
- Vibration window analysis.

- Audio window analysis.

4.4.2 Health engine

- Automatic baseline calibration.
- Deterministic machine-health engine.
- Temporal smoothing and hysteresis.
- Missing-modality handling.

4.4.3 Telemetry and persistence

- Live WebSocket telemetry to the dashboard.
- Event and history persistence in SQLite.

4.4.4 AI and signal processing

- ML models: MobileNetV3-Small (thermal, transfer learning), Random Forest (vibration) and Logistic Regression (acoustic, 61 features).
- Analysis and features: FFT / STFT, octave / 1/3-octave bands and MFCC.
- Anomaly and fusion: statistical z-score with Isolation Forest, confidence gating and weighted ML evidence fusion.

4.5 User Interface

The current dashboard is a single-page interface with input selection, a test window and an output area. The user selects an available dataset or video and starts or stops a test. The output presents model predictions and confidence where returned by the backend, along with known dataset labels where available. Video playback and backend frame analysis may progress at different rates. The dashboard does not present a validated fused diagnosis for recordings that are not verified as synchronized.

4.6 Implementation Workflow

- Select a machine and its thermal, vibration and acoustic dataset sources.
- Stream data to the edge pipeline for preprocessing and feature extraction.
- Run the deterministic health engine and the modality-specific ML models.
- Apply confidence gating and fuse the available evidence.
- Publish health score, state and evidence over WebSocket.
- Persist events and history and display them on the dashboard.

CHAPTER 5

RESULTS AND DISCUSSION

5. RESULTS AND DISCUSSION

5.1 Experimental Setup

The prototype was evaluated in a dataset-driven configuration: thermal video, vibration signals and acoustic recordings from public datasets simulate real sensor inputs, and the full pipeline (acquisition, processing, inference, fusion and dashboard) runs end-to-end on an edge-style computer. A demonstration machine set of seven usable videos covering six machine types was used.

Table 5.1: Datasets Used in the Prototype

Dataset	Modality	Source
IMS Bearing Run-to-Failure Dataset	Vibration	NASA / University of Cincinnati (see [1])
Induction Motor - Thermal Images Dataset	Thermal	Mendeley Data (see [2])
Thermal Image of Equipment + 40 Ground Truths	Thermal	Mendeley Data (see [3])
DCASE 2022 Task 2 - Machine Condition Monitoring (bearing acoustic)	Acoustic	DCASE Challenge (see [4])

5.2 Prototype Results

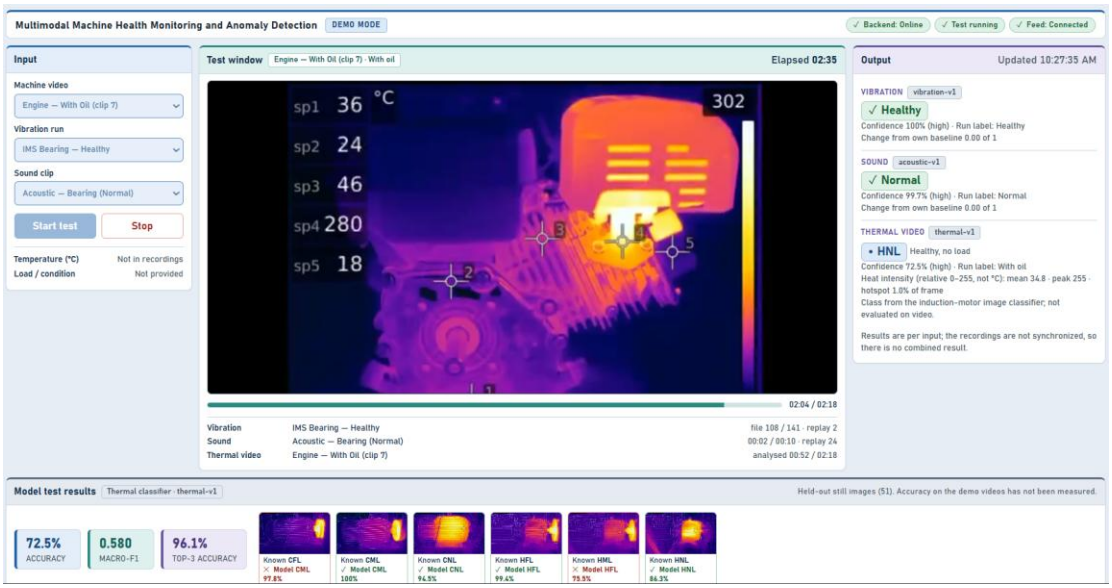


Figure 5.1: Monitoring Dashboard (current single-page interface)

The dashboard demonstrates a thermal feed of an engine with oil, with hotspot-intensity readings (mean, maximum, minimum, hotspot threshold, hotspot area and temperature difference). The output presents modality-level predictions and confidence where returned by the backend. The system reports the machine state (for example NORMAL)

along with a health history. The demonstration is a prerecorded video, not a live physical thermal camera.

Table 5.2: Key Results of the Prototype

Result	Value	Details
Sensing modalities	3	Thermal, vibration, acoustic
Machine-learning models	3	CNN (MobileNetV3-Small), Random Forest, Logistic Regression
Backend tests passed	194	Unit, integration and API tests passed in the reported test run
Demonstration videos	7	Usable videos in the demo machine set
Machine types	6	Honda engine (V1, V2), hydraulic system, EV battery, transformer, engine with oil, engine without oil

Table 5.3: Demonstration Machine Set

Machine	Notes
Honda engine (V1, V2)	Two usable videos
Hydraulic system	One usable video
EV battery	One usable video
Transformer	One usable video
Engine with oil	One usable video
Engine without oil	One usable video

5.3 Capabilities Demonstrated

- Thermal video-based inference (demonstration).
- Vibration signal analysis.
- Acoustic signal analysis.
- Modality-specific ML inference.
- Confidence-gated predictions.
- Weighted evidence-fusion component (not yet validated on synchronized recordings).
- Real-time health dashboard on recorded inputs.
- Event and health-history tracking.

5.4 Discussion

The prototype demonstrates the thermal, vibration and acoustic processing paths using recorded datasets. The dashboard presents the available modality-level results. The recordings are not verified as synchronized measurements of the same machine, so the

demonstration does not establish a validated combined diagnosis or prove that fusion improves performance.

Keeping the deterministic health engine authoritative and using ML outputs as supplementary evidence keeps the machine-health indication explainable. Confidence gating is designed to reduce false alerts from uncertain model outputs. Missing-modality handling allows the software to continue operating when one data stream is unavailable, which matters for real deployments where sensors may fail or be absent.

The passing of 194 backend unit, integration and API tests in the reported run supports the reliability of the software pipeline. Processing runs locally on the development computer; response times on the target edge hardware have not yet been measured.

5.5 Limitations

- The prototype currently uses prerecorded datasets rather than live physical sensors.
- Quantitative results are available for selected model evaluations but are limited by dataset size and label quality. The thermal image evaluation reports accuracy of 0.7255, macro-F1 of 0.5799 and top-three accuracy of 0.9608 on 51 held-out images. These are image-test results, not a measured video-accuracy result. The vibration model's labels are derived partly from RMS and kurtosis, which are also model inputs, so its 1.0 F1 result is circular; the spectral-only F1 of 0.857 is the more informative estimate. The acoustic detector flags 20% of normal files in the reported evaluation. Latency measurements on physical hardware are not yet reported.
- The thermal model was evaluated on held-out thermal images from its training dataset. Its predictions on the demonstration videos are not covered by that evaluation and may be out of domain; video accuracy has not been measured.
- Vibration and acoustic models use classical ML (Random Forest, Logistic Regression); specialised deep-learning models are planned.
- Remaining Useful Life estimation has not yet been implemented.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project delivers a dataset-driven prototype with separate thermal, vibration and acoustic analysis paths, a backend health-monitoring pipeline and a dashboard for viewing available results. The demonstration shows software inference and telemetry on recorded inputs. It does not establish a validated diagnosis across synchronized modalities, remaining useful life, prevention of equipment failures or performance on a production machine.

Key outcomes

- Three sensing modalities: thermal, vibration and acoustic.
- Three modality-specific ML models: MobileNetV3-Small, Random Forest and Logistic Regression.
- Confidence-gated predictions to suppress uncertain ML outputs.
- Weighted evidence-fusion component with missing-modality handling (demonstration only).

6.2 Innovation

- Modality intelligence: preserves domain signal features before ML.
- Evidence-level fusion: combines normalised evidence where available, without forcing unaligned raw feeds; not yet validated on synchronized data.
- Confidence gating: designed to suppress uncertain model inferences and reduce alert fatigue.

6.3 Impact

The prototype demonstrates how machine-condition evidence can be presented in one interface. Its potential to support maintenance decisions, reduce downtime or operate with real-time performance requires evaluation using synchronized data from physical machines and measured latency on the target hardware. Any 0-100 health score is a score, not a percentage probability.

- Local processing: the prototype runs its processing locally on the development computer. Raspberry Pi deployment and response-time performance on that hardware remain to be verified.

6.4 Deliverables

- Backend server: FastAPI + WebSocket real-time analytics engine.
- Monitoring dashboard: React and TypeScript single-page interface for selecting test inputs, viewing video playback and inspecting backend model outputs.
- AI pipeline: edge-optimised MobileNetV3 + Isolation Forest modules.

6.5 Future Scope

- Specialised models: dedicated vibration and acoustic deep-learning models.
- RUL estimation: Remaining Useful Life regression based on health trajectory.
- Synchronized multimodal data: collect thermal, vibration and acoustic recordings from the same machine at the same time to test fusion.
- Plant pilot: direct hardware DAQ interfacing with shop-floor motors.

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Prof. Ashwini Lokhande (Signature)	Dr.Rashmi Welekar	Dr. P. D. Adane

ANNEXURE A: PUBLICATIONS AND INTELLECTUAL PROPERTY

This annexure should include documentary evidence related to project outcomes such as published or submitted papers, conference publications, patent applications, design registrations and copyright registrations. At the time of this report, no publications or intellectual property filings have been recorded.

Type	Title / Description	Status
Research Paper	--	Not yet submitted
Patent / Design	--	Not filed
Conference Paper	--	Not yet submitted

ANNEXURE B: STUDENT CONTRIBUTION STATEMENT

Each student should specify a specific and verifiable contribution. Contribution entries below are to be completed by the team before submission.

Student Name	Roll Number	Contribution
Vaishnav Balpande	B1-01	Thermal analysis: MobileNetV3-Small transfer learning, held-out thermal image evaluation (accuracy, macro-F1, top-three accuracy) and hotspot feature extraction.
Shreyank Nipane	B4-55	Vibration analysis: windowed RMS, kurtosis, crest factor and FFT features; Random Forest model; IMS dataset preparation; spectral-only F1 evaluation.
Tanushri Sapate	B3-44	Acoustic analysis: STFT, octave-band and MFCC features; Logistic Regression on 61 features; false-alarm evaluation on normal recordings; DCASE dataset preparation.
Aastha Pusdekar	B4-52	Backend and health engine: FastAPI and WebSocket telemetry, SQLite persistence, z-score / Isolation Forest / hysteresis health engine, backend test suite (194 tests).
Kartik Madankar	B3-38	Dashboard and report: React/TypeScript single-page dashboard, video playback and backend output display, figure preparation, literature review and report writing.

ANNEXURE C: PROJECT OUTPUTS

Major outputs generated from the project include:

- Working prototype: multimodal machine-health monitoring platform.
- Software / application: FastAPI + WebSocket backend and React + TypeScript + Vite operator dashboard.
- AI pipeline: MobileNetV3-Small (thermal), Random Forest (vibration) and Logistic Regression (acoustic) models with Isolation Forest anomaly detection.
- Technology demonstration: demo machine set of seven usable videos across six machine types.
- Test suite: 194 backend unit, integration and API tests in the reported run.